Master of Business Analytics

Prescriptive Analytics & Optimization

MBAN 5540

Week 02
LP Graphical Solution
Excel Solver

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LP Review

- The process of finding the best value (Minimum/Maximum) for an objective function by changing decision variables with respect to constraints.

➢ Decision Variables:

- Decision choices that are under control, such as the number of units of a product to produce.

➢ Objective Function:

- The goal of the problem, such as maximizing profit or minimizing cost

➢ Constraints:

- A set of restrictions or limitations that prevent us for selecting any value we please for the decision variables, such as production capacities.

Flair Company LP Model

$$Maximize \quad 7T + 5C$$

Objective Function

Subject to:

Constraints

$$3T + 4C \leq 2{,}400$$
$$2T + 1C \leq 1{,}000$$
$$C \leq 450$$
$$T \geq 100$$
$$T \geq 0$$
$$C \geq 0$$

Non-negativity Constraints

Decision Variables

T: Number of tables produced
C: Number of chairs produced

Solving LP

- Graphical Solving
  - For problems with at most 2 variables
  - Very limited
  - Good illustration for educational purposes

- Algorithms:
  - SIMPLEX
  - Interior-point methods
  - …
- Computer packages
  - CPLEX
  - Gurobi
  - GAMS
  - Excel Solver

Graphical Solution Steps

1. Draw all the constraints to identify the feasible region.

   - Feasible region: Points that satisfy ALL the constraints

   - Feasible Solution: Any point in the feasible region is a feasible solution.

Drawing Constraints

- Each inequality constraint identifies a half-plane.

- Steps:

  1. Draw the line corresponding to the equation (not inequality) by finding two points.

  2. Pick a random point (usually (0,0)) to test which half-plane is defined by the inequality

Graphical Solution Steps

It is guaranteed that the optimal solution is always one of the corner points.

2. Find the xy - coordinate of all the corner points.

3. Put each corner point's coordinate into the objective function and calculate its value. The point that maximizes (minimizes) the objective function is the optimal solution.

Optimal Solution: The best (with respect to the objective function) point of the feasible solution (which is always on the corner!)

Drawing Objective Function

As an ALTERNATIVE to finding the xy-coordinate of all corner points, we can draw the objective function contour (level, ISO) lines to find the optimal solution.

- Steps:
  - Set the objective function equal to a random number (Choose wisely: Least Common Multiple (LCM)!). Then draw the line.
  - Contour lines are all parallel to the one you draw.
  - Move the contour line (according to objective!) to leave the feasible region. The last corner node is the <u>optimal solution</u>.

Graphical Solution Notes

- Always shade in the feasible region.
- Label all feasible corner points and determine their coordinates.
- Find the optimal corner (optimal solution) and the corresponding objective function value.

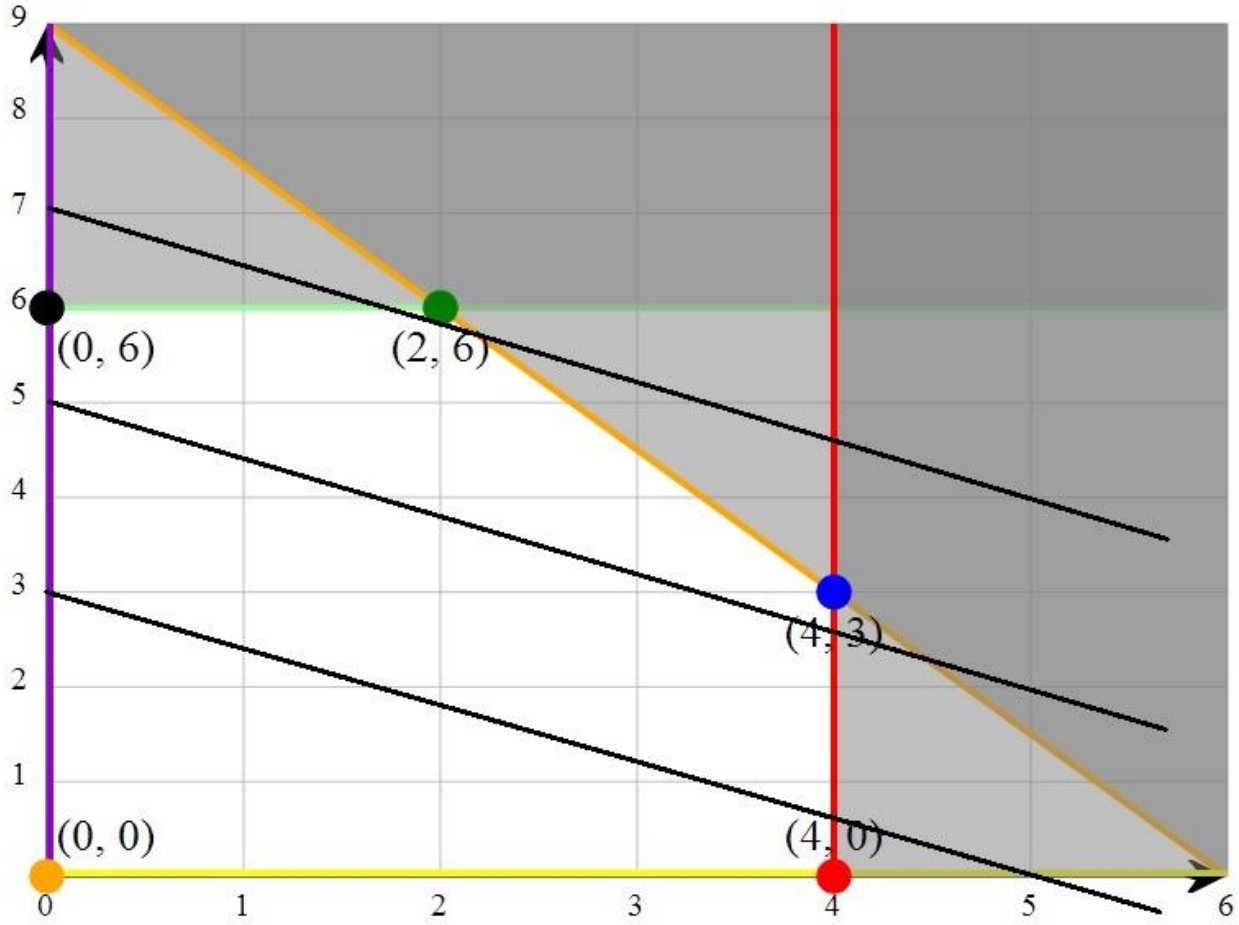
Graphical Solution: Ex1

$$Maximize \quad 3x + 5y$$

$$3x + 2y \leq 18$$
$$x \leq 4$$
$$2y \leq 12$$
$$x \geq 0$$
$$y \geq 0$$

Ex1: Optimal Solution

Ex1: Optimal Solution



$(x, y) = (2,6)$ is the optimal solution.

Objective function value $= 3x + 5y = 36$

Example: Flair Company

- Produces tables and chairs

	Per Table	Per Chair	Availability
Profit ($)	7	5	
Carpentry (hours)	3	4	2,400
Painting (hours)	2	1	1,000

- Manager/Market restrictions:
  - Make no more than 450 chairs
  - Make at least 100 tables

Flair Company LP Model

$$Maximize \;\; 7T + 5C$$

$$3T + 4C \leq 2{,}400$$

$$2T + 1C \leq 1{,}000$$

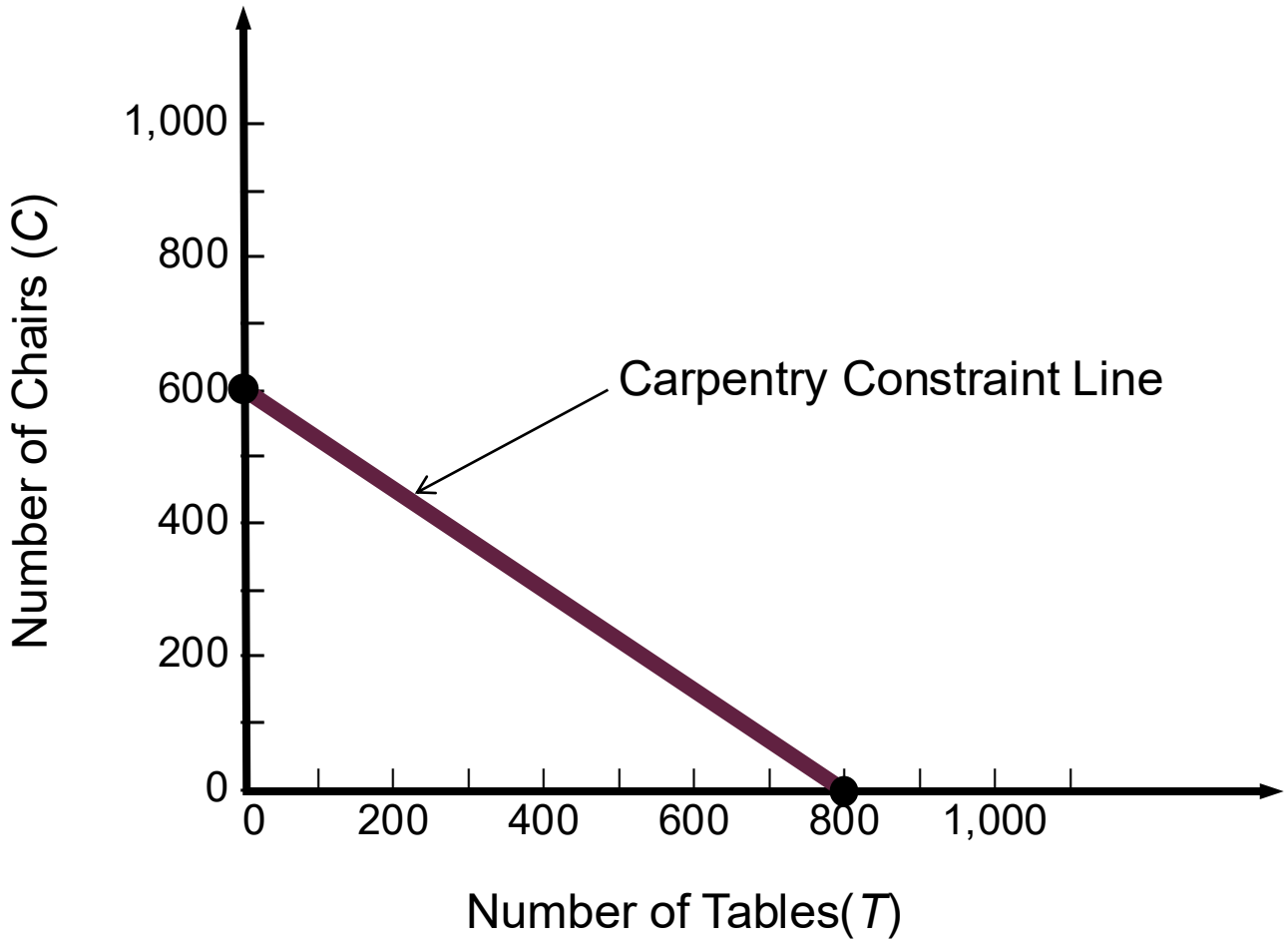
$$C \leq 450$$

$$T \geq 100$$

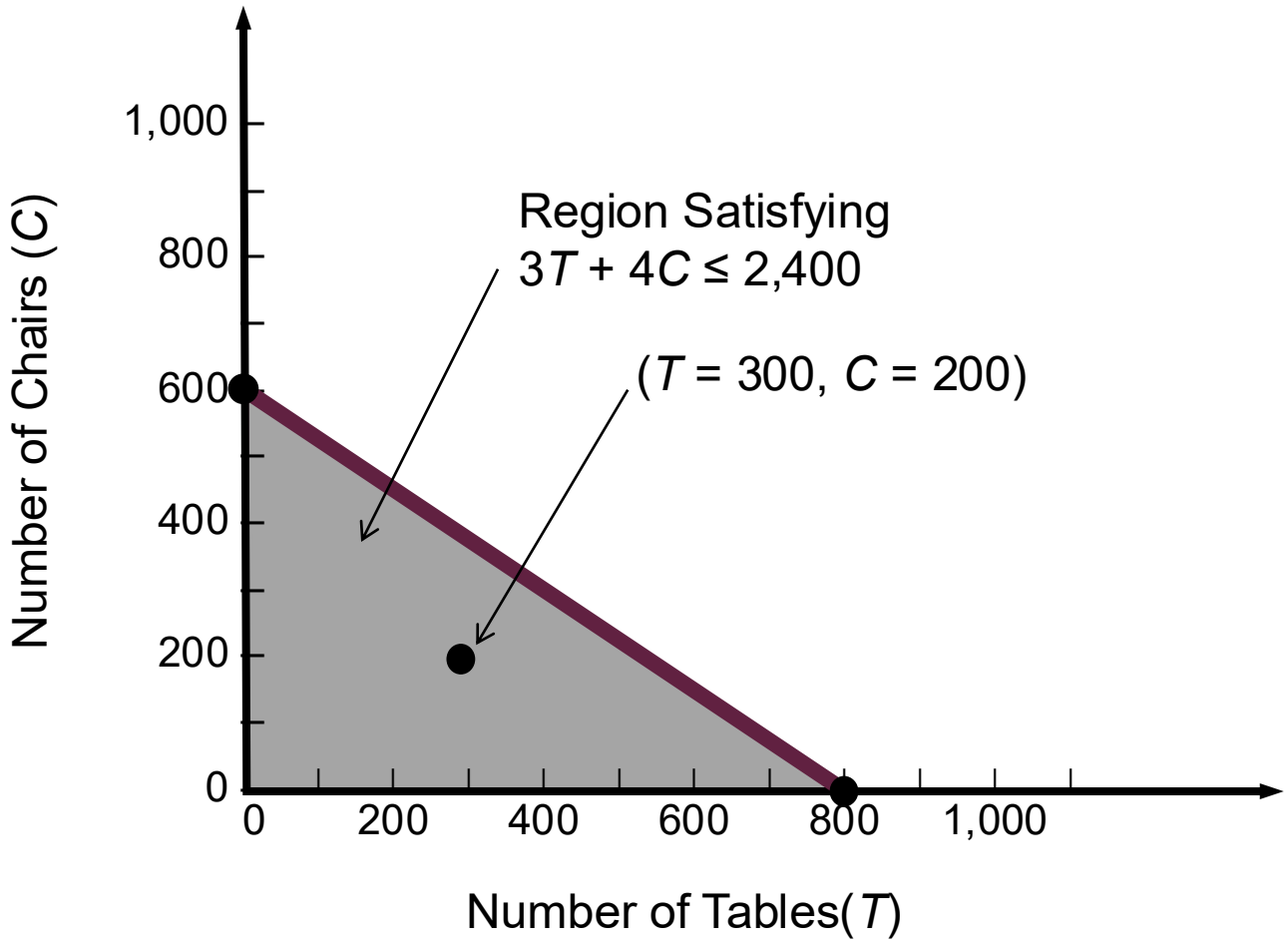
$$T \geq 0$$

$$C \geq 0$$

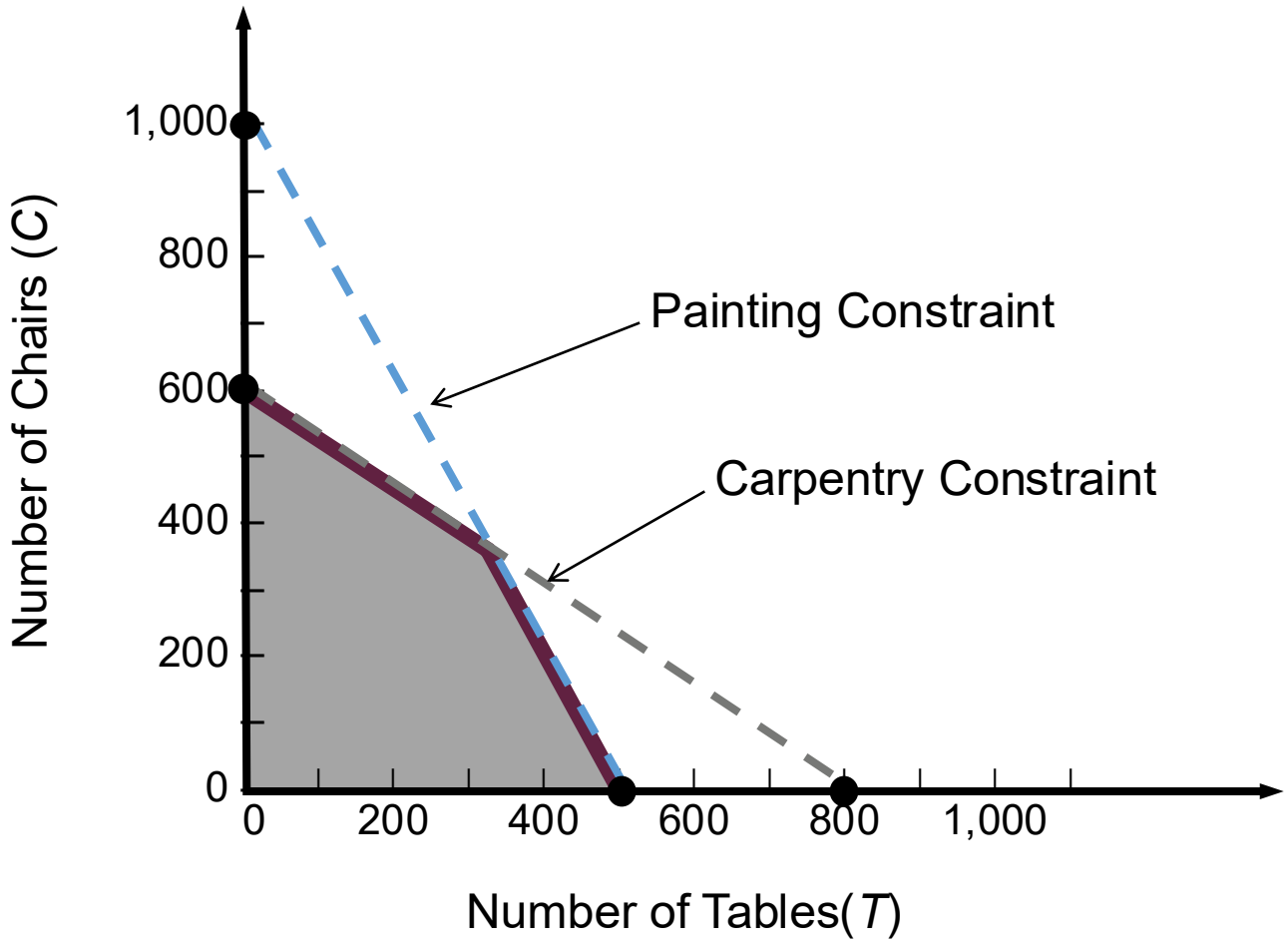
Flair Graphical LP Solution



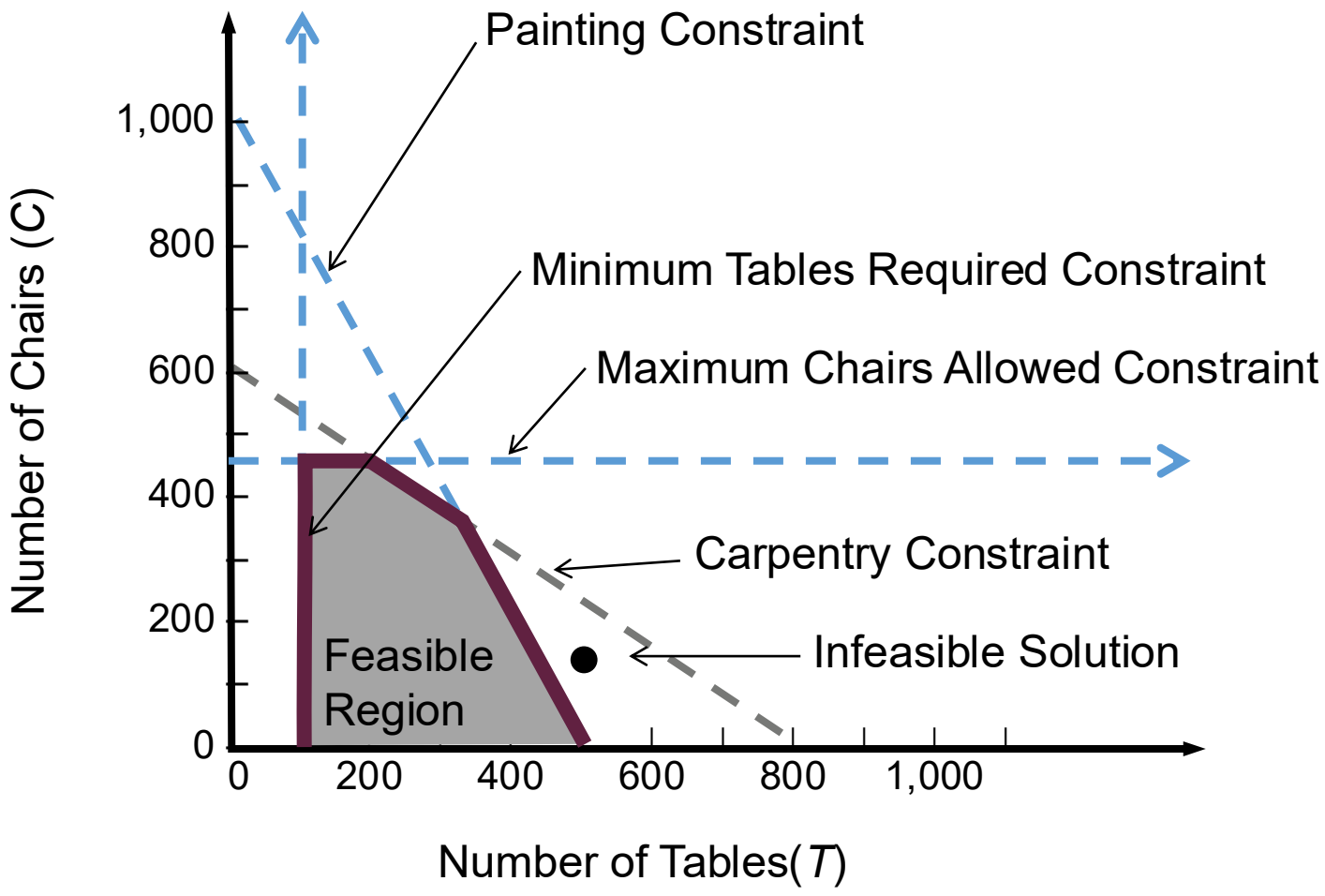
Flair Graphical LP Solution



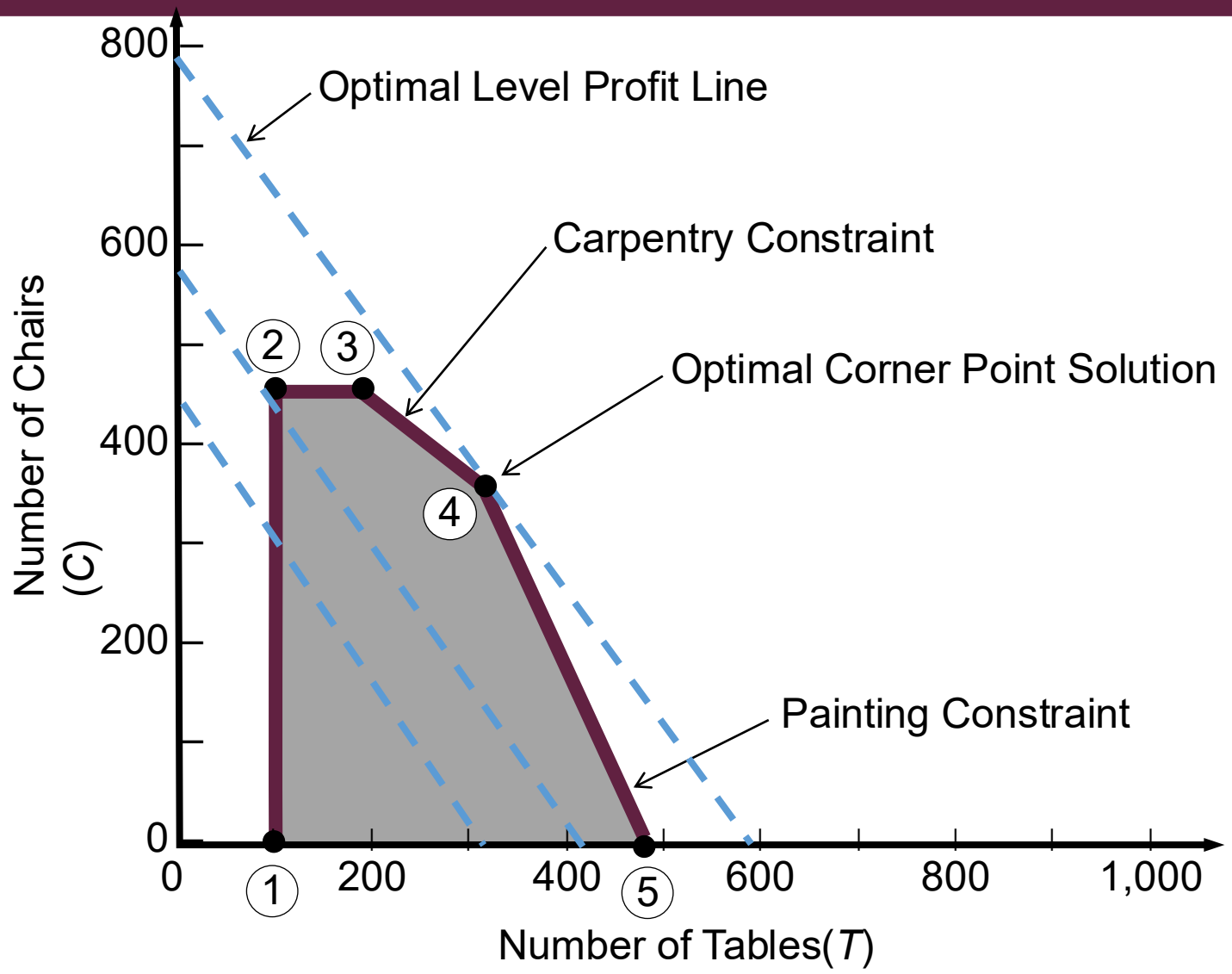
Flair Graphical LP Solution



Flair Graphical LP Solution



Flair Graphical LP Solution



Flair Graphical LP Solution

Point 1	(T = 100, C = 0)		
	Profit = \$7(100) + \$5(0)	=	\$700
Point 2	(T = 100, C = 450)		
	Profit = \$ 7(100) + \$5(450)	=	\$2,950
Point 3	(T = 200, C = 450)		
	Profit = \$ 7(200) + \$5(450)	=	\$3,650
Point 4	(T = 320, C = 360)		
	Profit = \$ 7(320) + \$5(360)	=	\$4,040
Point 5	(T = 500, C = 0)		
	Profit = \$ 7(500) + \$5(0)	=	\$3,500

Flair Graphical LP Solution

- Optimal solution:
- Intersection of carpentry constraint and painting constraint:

$$3T + 4C = 2{,}400$$
$$2T + 1C = 1{,}000$$

$$T = 320 \quad C = 360$$

- Objective function value:

$$7T + 5C = 7(320) + 5(360) = \$4{,}040$$

Special Situations in LP

1) Alternate Optimal Solution

2) Redundant Constraints

3) Unbounded Solutions

4) Infeasibility

1- Alternate Optimal Solution

- When there is more than one corner point that maximize or minimize the objective function

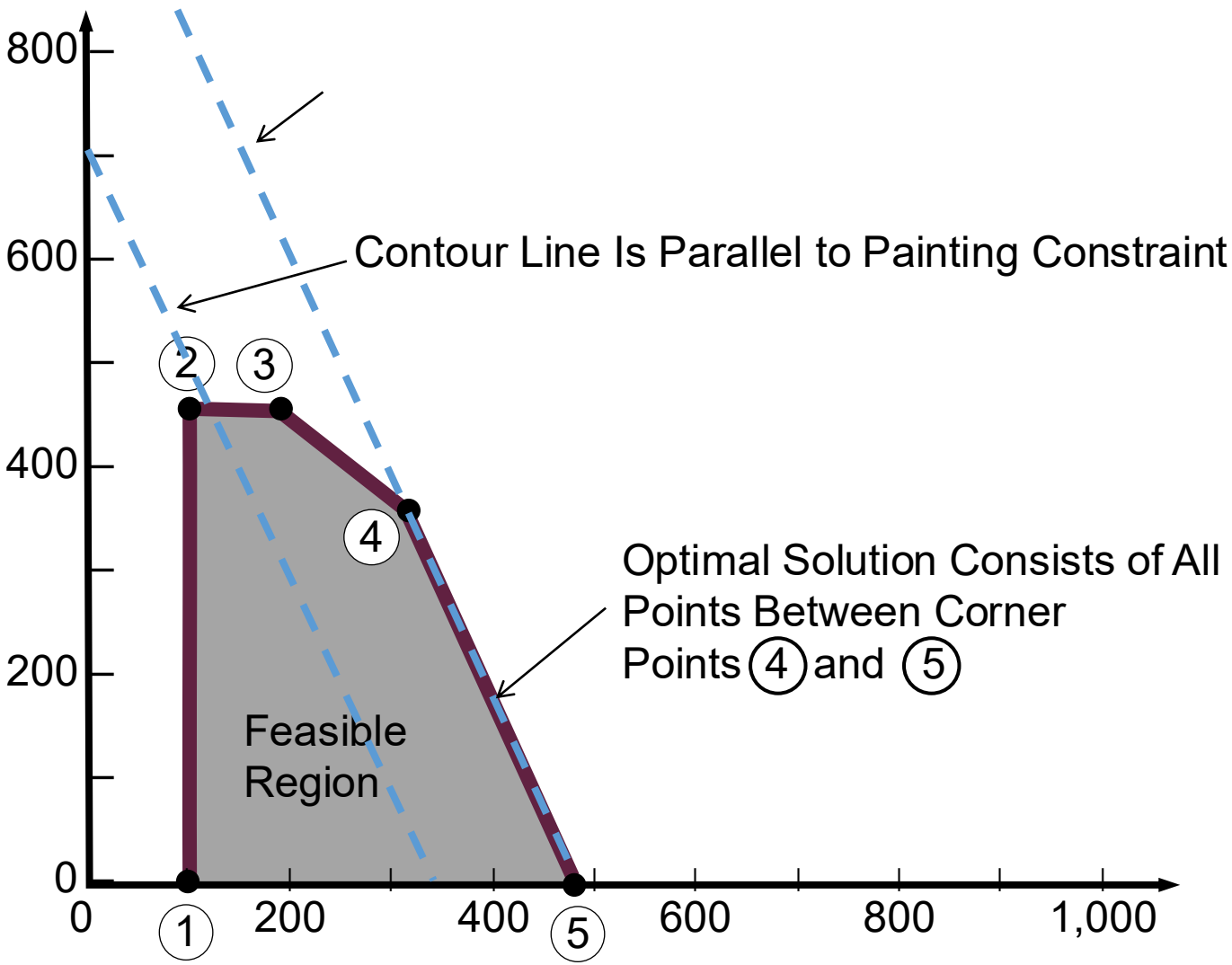
- Happens when the objective contour line is parallel to one of the constraints. (same slope)

$$
\begin{aligned}
Max\ 3x_1 + 4x_2 \\
x_2 &\leq 2 \\
6x_1 + 8x_2 &\leq 24 \\
x_1 &\leq 3 \\
x_1 &\geq 0 \\
x_2 &\geq 0
\end{aligned}
$$

$$\frac{3}{6} = \frac{4}{8}$$

$$Slopes = -\frac{3}{4}$$

1- Alternate Optimal Solution



2- Redundant Constraint

$$Max\ x_1 + x_2$$

$$x_1 + x_2 \leq 4$$

$$x_2 \leq 3$$

$$x_1 + x_2 \leq 6$$

$$x_2 \leq 5$$

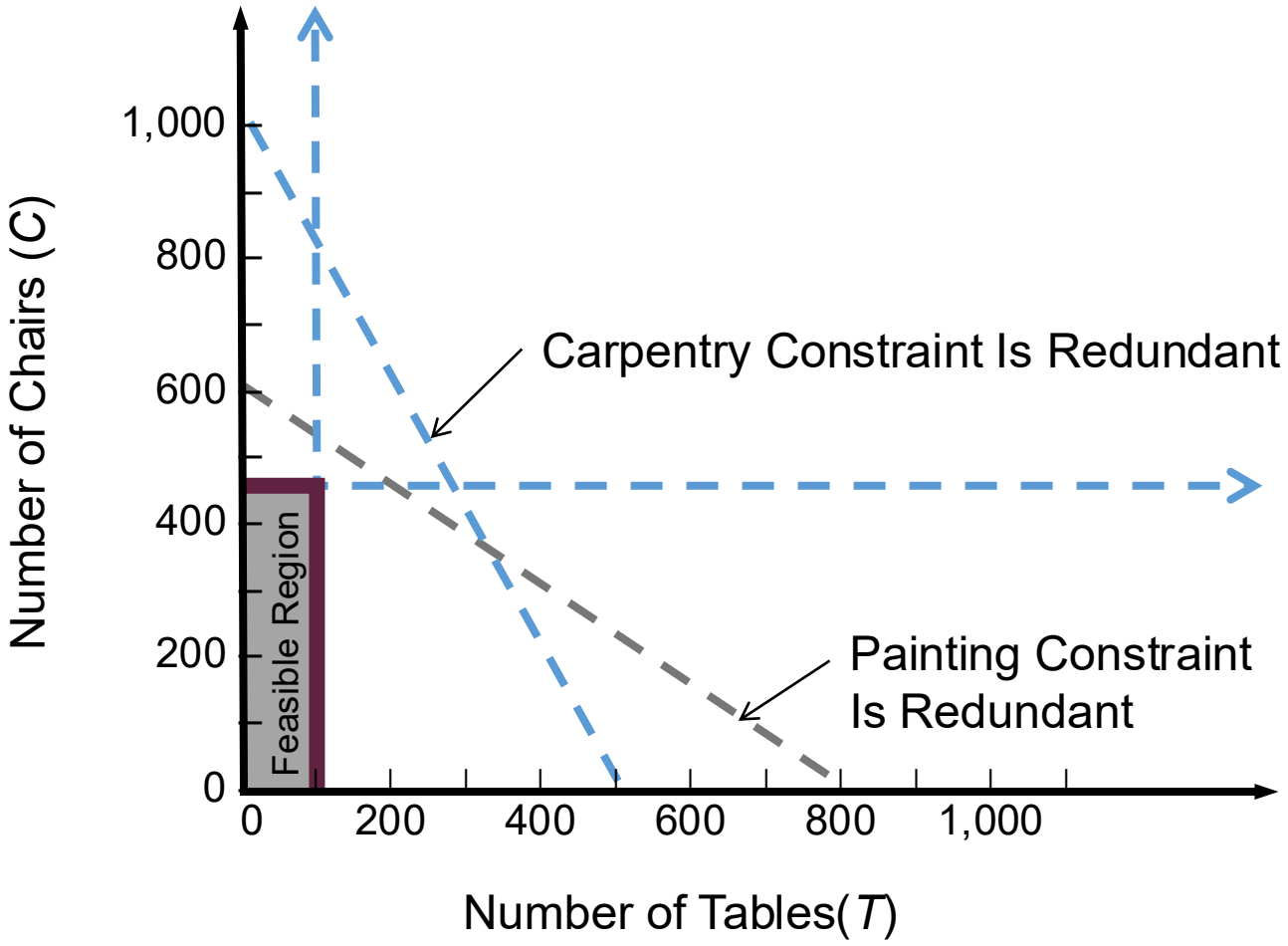
$$x_1 \geq 0$$

$$x_2 \geq 0$$

Redundant Constraints

2- Redundant Constraint

- A redundant constraint plays NO role in identifying the feasible region !



3- Unbounded Solution

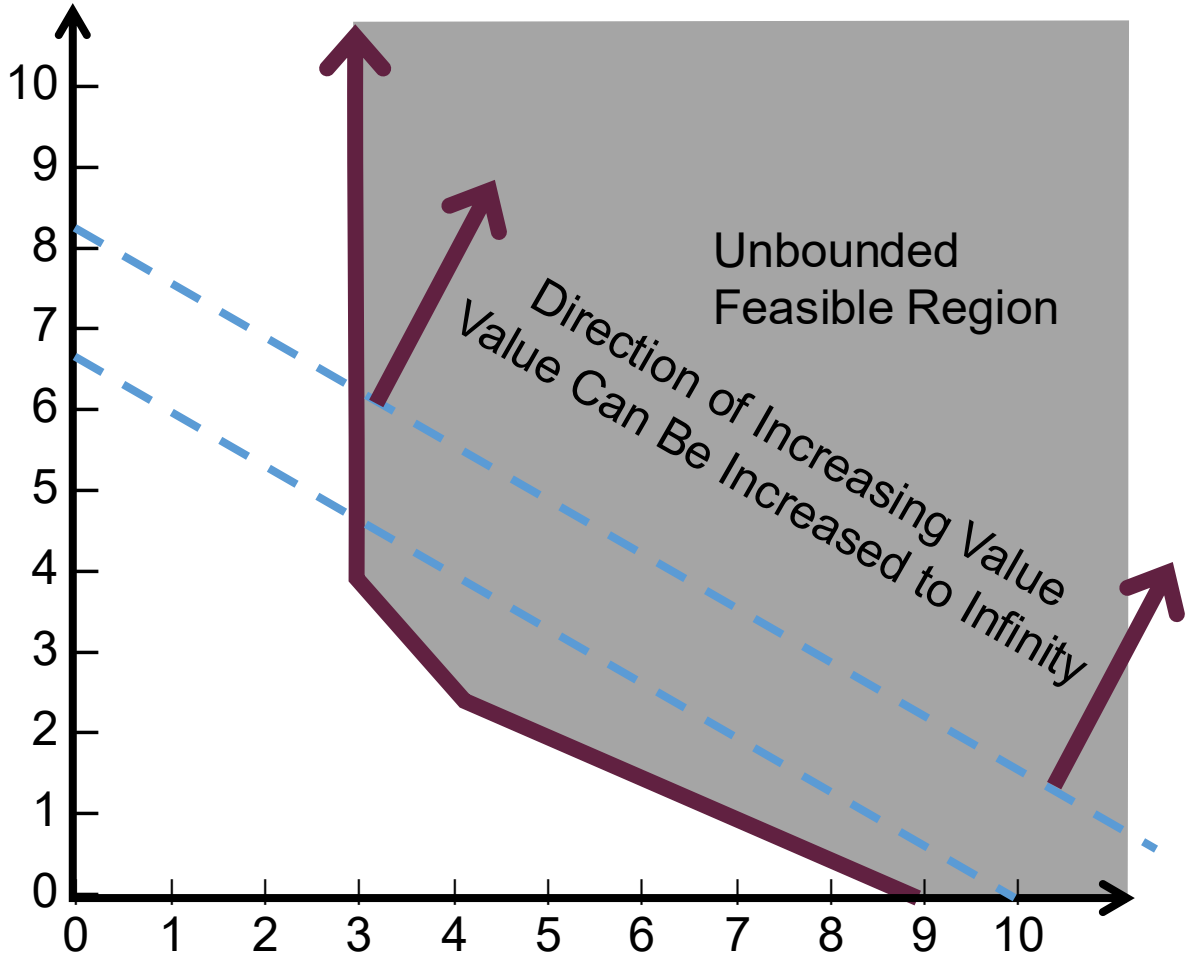
- When the objective function can be infinitely large (for MAX) or infinitely small (for Min)

$$
\begin{aligned}
Max\ x_1 + x_2 & \\
x_1 + x_2 &\geq 4 \\
-x_1 + x_2 &\leq 4 \\
x_1 &\geq 0 \\
x_2 &\geq 0
\end{aligned}
$$

Note: It depends on the objective function.

Unbounded Solution VS. Unbounded Feasible Region

3- Unbounded Solution



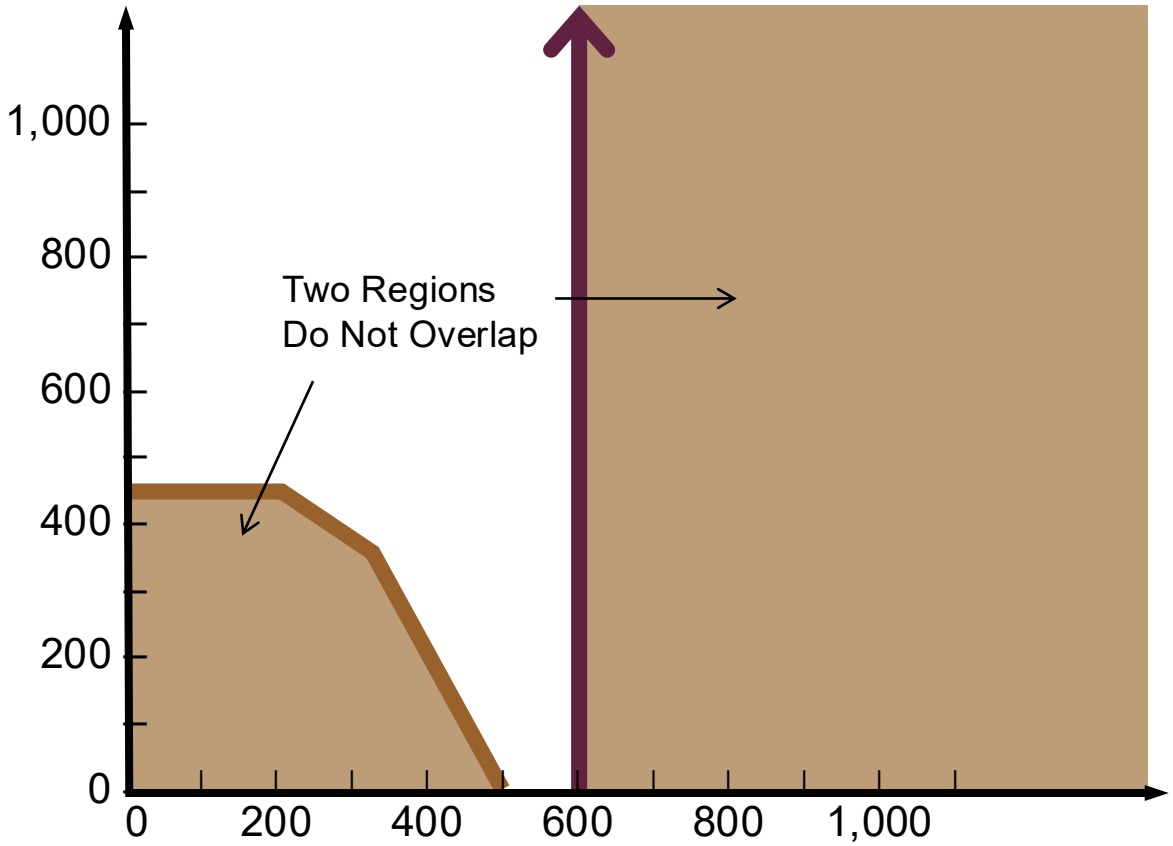
4- Infeasibility

- An LP is infeasible if all constraints cannot be satisfied at the same time.

$$
\begin{aligned}
Max\ x_1 + x_2 \\
x_1 + x_2 &\leq 3 \\
x_1 + x_2 &\geq 5 \\
x_1 &\geq 0 \\
x_2 &\geq 0
\end{aligned}
$$

- No Solution in this case !

4- Infeasibility



Modeling in Microsoft Excel

- Microsoft Excel can be used to model and solve a Linear Programming problem.
- Solving the LP in Excel requires an Add-in called “Solver”.
- Excel Solver comes with the basic installation of Microsoft Excel, but it needs to be activated.

Activating Solver in Microsoft Excel

Microsoft Windows

1) Click ‘File’ and then the ‘Options’.
2) Go to the ‘Add-Ins’ command.
3) At the bottom, select “Excel Add-ins” for Manage and hit “Go…”.
4) The ‘Add-Ins’ dialog box will appear. Check ‘Solver Add-In’; click the ‘OK’.
5) ‘Solver’ is now loaded and is available in the Excel ‘Data’ tab.

Mac OS

1) Open Excel for Mac.
2) Go to the Tools menu, select "Add-ins“
3) Check "Solver Add-in“, then click OK .
4) Select the "Data" tab, then select "Solver".

Modeling in Microsoft Excel

- We need to identify the below items for Solver:
  - In which cells are the values of the **decision variables** that it can change. Excel calls these the **Changing Variable Cells**.
  - In what cell is the value of the **objective function** (e.g., total profit). Excel calls this the **Set Objective**.
  - Do you want to **Max**imize the target Cell, or **Min**imize it?
  - What are the **Constraints**?

Recall: Flair Company LP Model

$$Maximize \;\; 7T + 5C$$

$$3T + 4C \leq 2{,}400$$

$$2T + 1C \leq 1{,}000$$

$$C \leq 450$$

$$T \geq 100$$

$$T \geq 0$$

$$C \geq 0$$

Modeling Example in Microsoft Excel

Define the layout of your model: Define the variables, objective function and constraints of your problem in separate rows.

H8

	A	B	C	D	E	F
1						
2						
3		Table	Chairs			
4		T	C			
5	Variables: # of units					
6	Objective: Profit					
7	Constraints					
8	Carpentry					
9	Painting					
10	Max Chair					
11	Min Table					
12						

Use of SUMPRODUCT Function

- The objective function are simple linear formulas composed of the product of a coefficient and a variable (e.g., 7T and 5C) and then these products are added (summed) together (e.g., 7T + 5C).

- In Excel this is most easily done using the **Sumproduct** function. This function needs to know where to find the coefficients and where to find the corresponding variables.

- For the total Profit, the coefficients are in B6 and C6. The variables are in B5 and C5. So, the total profit **=Sumproduct(B6:C6,B5:C5)**. If there were more variables and coefficients in columns D, E and F, say, then we would have sumproduct(B6:F6,B5:F5). By using the **:** between the first and last cell in the list, Excel knows to include all those in between.

- The decision variables cells will not change, and we want to use (copy and paste formula) the cells of decision variables for defining the constraints. We can force this by putting a **$** sign in front of the decision variable cell's row and column labels. That is, **=sumproduct(B6:C6,$B$5:$C$5)**.

Use of SUMPRODUCT Function

D19

	A	B	C	D	E	F
1						
2						
3		Table	Chairs			
4		T	C			
5	Variables: # of units					
6	Objective: Profit	7	5	=SUMPRODUCT(B6:C6,$B$5:$C$5)		
7	Constraints					
8	Carpentry	3	4			
9	Painting	2	1			
10	Max Chair		1			
11	Min Table	1				
12						

D6 =SUMPRODUCT(B6:C6,$B$5:$C$5)

	A	B	C	D	E	F
1						
2						
3		Table	Chairs			
4		T	C			
5	Variables: # of units					
6	Objective: Profit	7	5	0		
7	Constraints					
8	Carpentry	3	4			
9	Painting	2	1			
10	Max Chair		1			
11	Min Table	1				
12						

Use of SUMPRODUCT Function

- The constraints are composed of three parts, the **LHS** (left hand side) total, the sign of the inequality (constraint), and the **RHS** (right hand side) limit. The LHS is in column D and the RHS is in column F.

- The LHS formula for carpentry constraint can be defined like objective function. So, the LHS=**sumproduct(B8:C8, $B$5:$C$5).** Therefore, the LHS formula for the constraints can similarly be copied from cell D6 (objective function formula). Copy cell D6 and paste in cells D8 to D11.

- Enter the sign of each constraint in column E.
- Enter the RHS of each constraint in column F.

Adding the Constraints

D18 Formula Bar

	A	B	C	D	E	F
1						
2						
3		Table	Chairs			
4		T	C			
5	Variables					
6	Objective	7	5	=SUMPRODUCT(B6:C6,$B$5:$C$5)		
7	Constraints					
8	Carpentry	3	4	=SUMPRODUCT(B8:C8,$B$5:$C$5)	<=	2400
9	Painting	2	1	=SUMPRODUCT(B9:C9,$B$5:$C$5)	<=	1000
10	Max Chair		1	=SUMPRODUCT(B10:C10,$B$5:$C$5)	<=	450
11	Min Table	1		=SUMPRODUCT(B11:C11,$B$5:$C$5)	>=	100
12				LHS	Sign	RHS
13						

D18

	A	B	C	D	E	F	G
1							
2							
3		Table	Chairs				
4		T	C				
5	Variables						
6	Objective	7	5	0			
7	Constraints						
8	Carpentry	3	4	0	<=	2400	
9	Painting	2	1	0	<=	1000	
10	Max Chair		1	0	<=	450	
11	Min Table	1		0	>=	100	
12				LHS	Sign	RHS	
13							

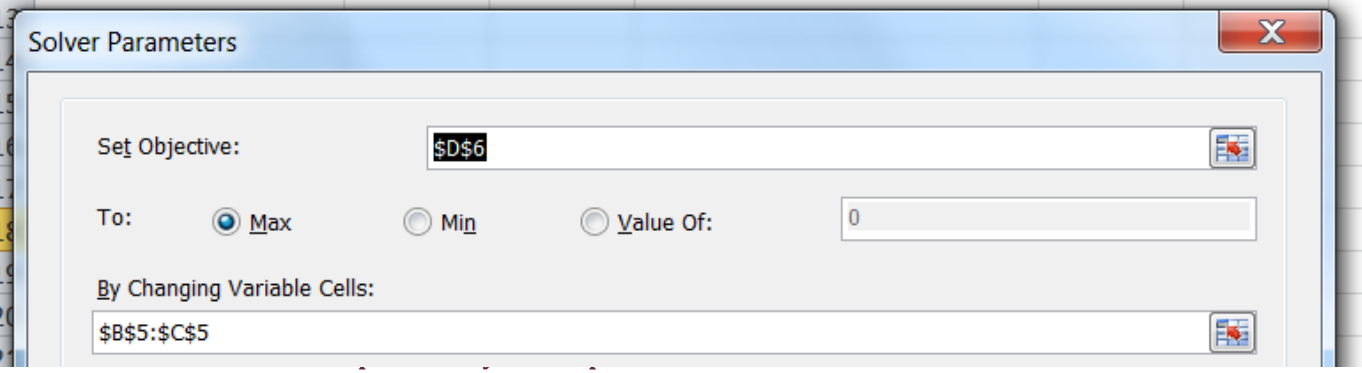
Modeling Example in Microsoft Excel

- The total profit is calculated in cell **D6**. This will be our objective or **Set Objective Cell**.
- We will want to find the **Max**
- The values of the variables will be in **B5** to **C5**. These will be our **Changing Cells**.
- The constraints are composed of three parts, the LHS (left hand side) total, the sign of the inequality (constraint), and the RHS (right hand side) limit. The LHS is in column D and the RHS is in column F.

Solver Inputs: Objective & Decision Variables

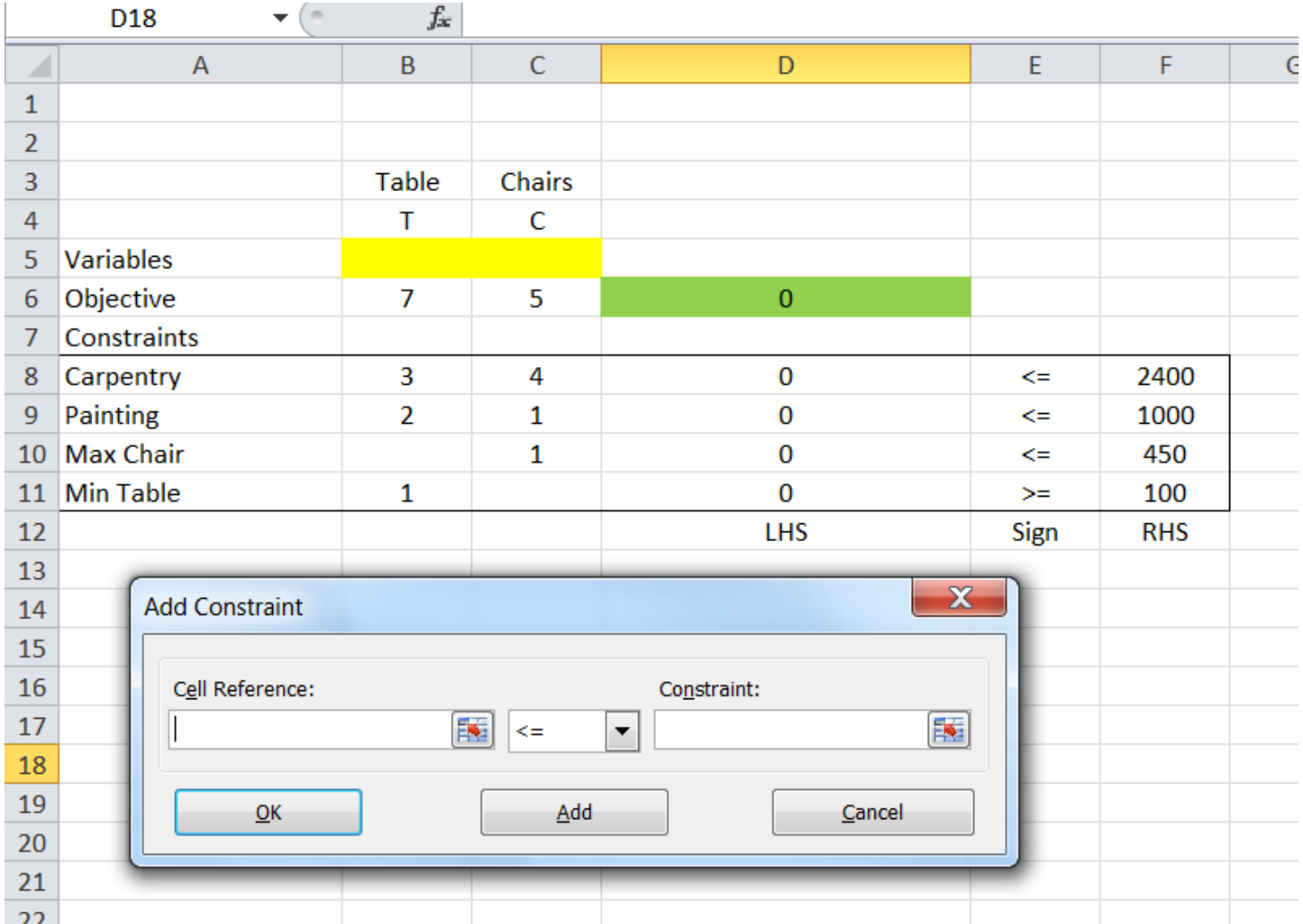
- Click on the Data tab and open Solver.
- For the Set Objective, click on D6. This is a maximizing question so make sure the Max button is selected. Click on the space marked By Changing Cells. Now select the cells B5 to C5 in your spreadsheet.

2						
3		Table	Chairs			
4		T	C			
5	Variables					
6	Objective	7	5	0		
7	Constraints					
8	Carpentry	3	4	0	<=	2400
9	Painting	2	1	0	<=	1000
10	Max Chair		1	0	<=	450
11	Min Table	1		0	>=	100
12				LHS	Sign	RHS



Solver Inputs: Constraints

- Click on the big empty space below **Subject to the constraints**. We need to **Add** constraints, so click the **Add** button. A new popup window appears.

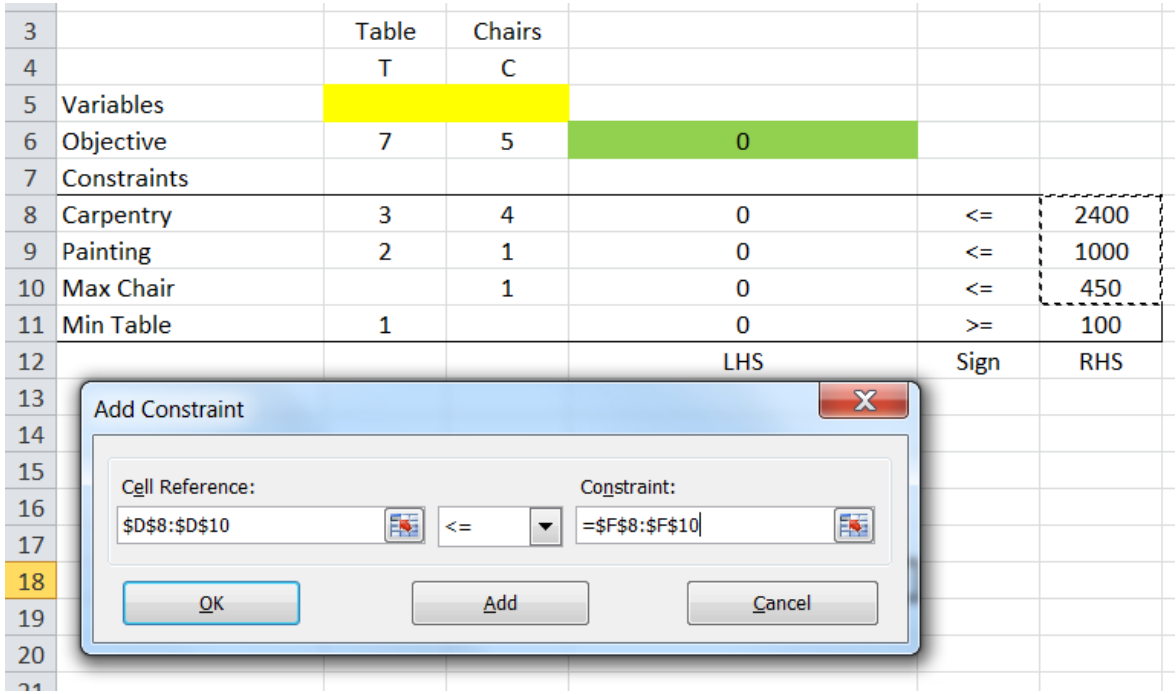


	A	B	C	D	E	F
1						
2						
3		Table	Chairs			
4		T	C			
5	Variables					
6	Objective	7	5	0		
7	Constraints					
8	Carpentry	3	4	0	<=	2400
9	Painting	2	1	0	<=	1000
10	Max Chair		1	0	<=	450
11	Min Table	1		0	>=	100
12				LHS	Sign	RHS

Solver Inputs: Constraints

- You can enter the constraints one by one or in groups. If you have several constraints together that all have the same sign (<= in the example), then you can enter them all at once. That's what we will do.

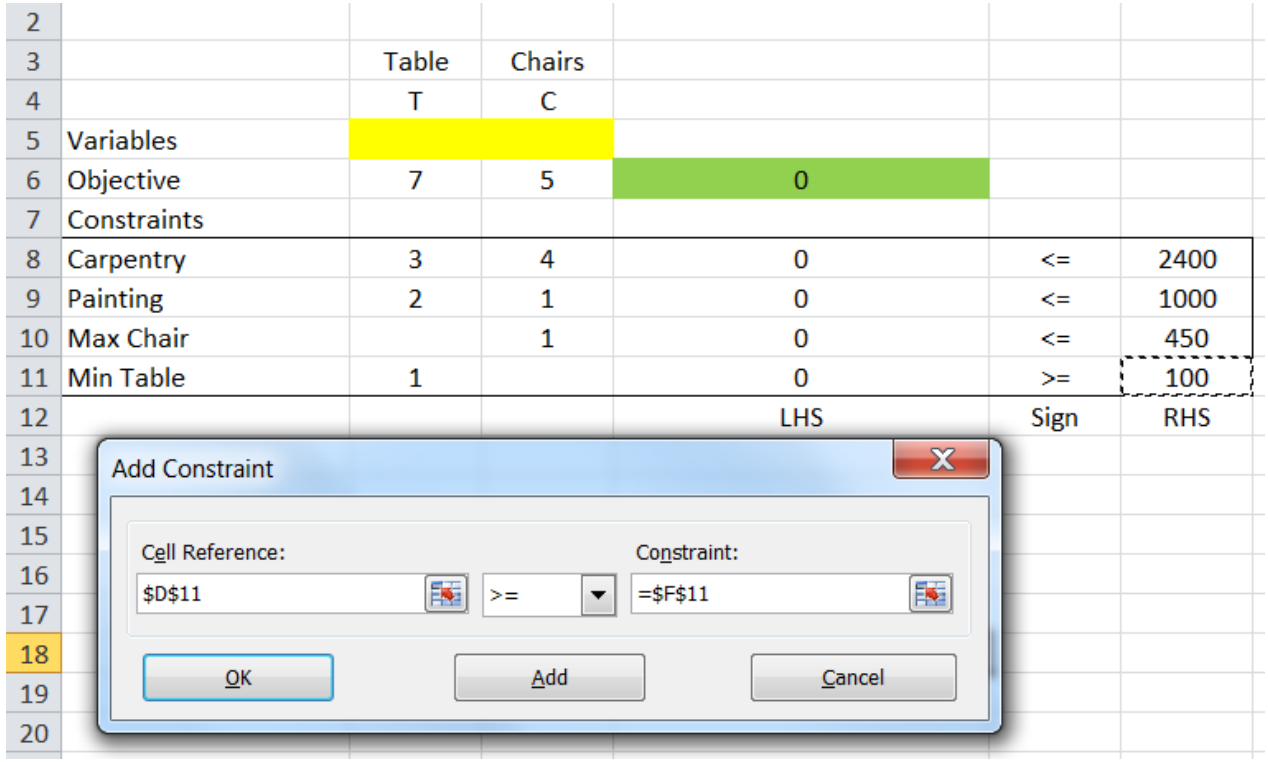
- Click on the **Cell Reference** box. Now click on the LHS totals (D8:D10). The inequality sign is already <=, but if we wanted to change it, we would click on the pull-down arrow and select the one we wanted. Click on the **Constraint** box and then click on the RHS values (F8:F10). Now click **OK**.



3		Table	Chairs			
4		T	C			
5	Variables					
6	Objective	7	5	0		
7	Constraints					
8	Carpentry	3	4	0	<=	2400
9	Painting	2	1	0	<=	1000
10	Max Chair		1	0	<=	450
11	Min Table	1		0	>=	100
12				LHS	Sign	RHS

Solver Inputs: Constraints

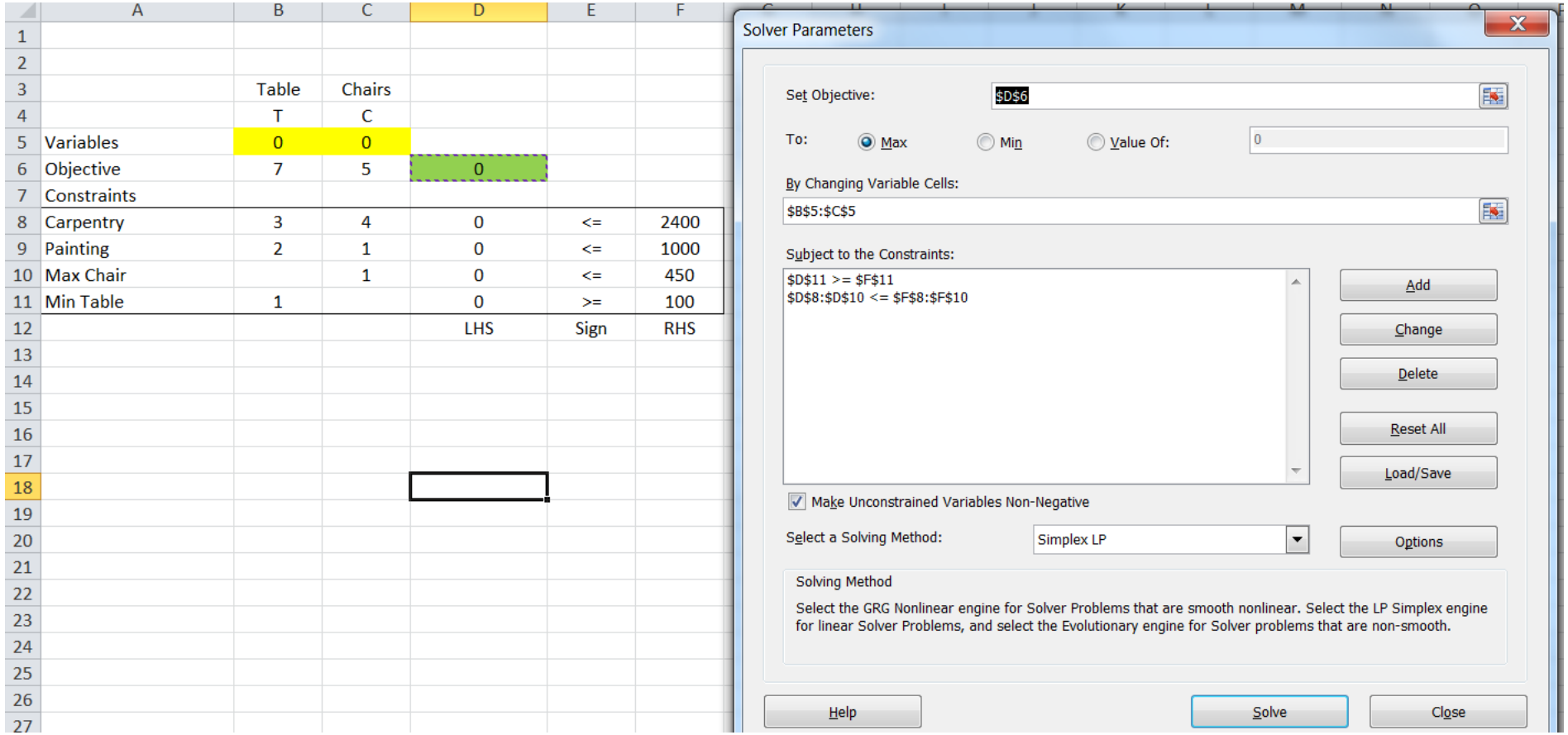
- Now we need to enter the Min Table constraint with sign >=. Click on the **Cell Reference** box. Now click on the LHS total (D11). The inequality sign is changed to >=. Click on the Constraint box and then click on the RHS value (F11). Now click OK.



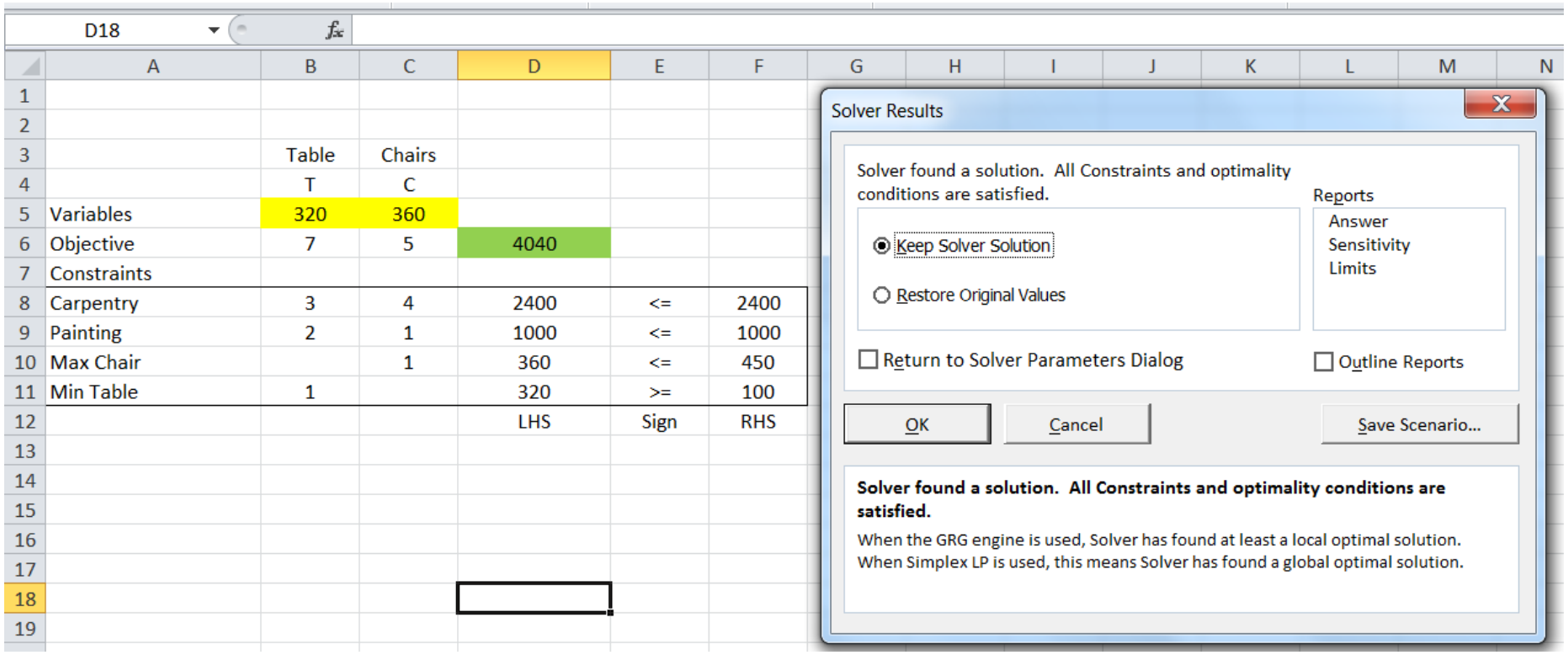
2						
3		Table	Chairs			
4		T	C			
5	Variables					
6	Objective	7	5	0		
7	Constraints					
8	Carpentry	3	4	0	<=	2400
9	Painting	2	1	0	<=	1000
10	Max Chair		1	0	<=	450
11	Min Table	1		0	>=	100
12				LHS	Sign	RHS
13						
14						
15						
16						
17						
18						
19						
20						

Solver Inputs: Non-negativity Constraints

- Check the box "**Make unconstrained variables Non-negative**" to enforce the no-negativity constraints of the decision variables.
- Choose "**Simplex LP**" (default) for solving all linear programming models.



	A	B	C	D	E	F
1						
2						
3		Table	Chairs			
4		T	C			
5	Variables	0	0			
6	Objective	7	5	0		
7	Constraints					
8	Carpentry	3	4	0	<=	2400
9	Painting	2	1	0	<=	1000
10	Max Chair		1	0	<=	450
11	Min Table	1		0	>=	100
12				LHS	Sign	RHS

Solver Solution

- Now click **Solve.** Solver will fill in the values of the variables and update the LHS values.
- Before looking at the results, it is important to read the message in the **Solver Results** window to verify that solver found an optimal solution.